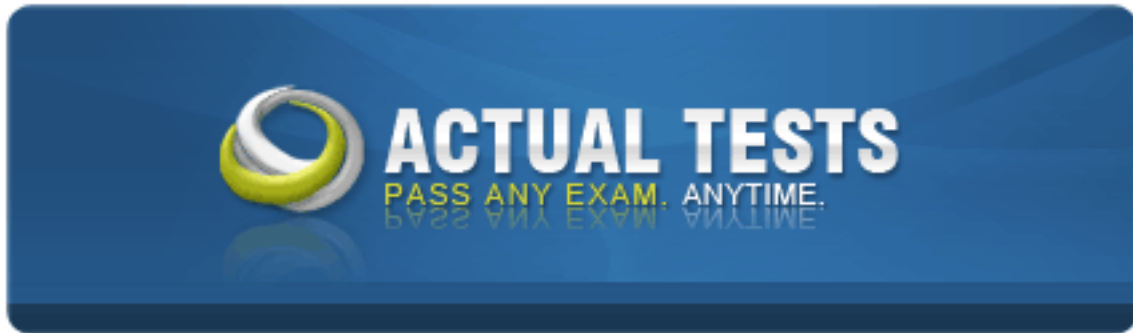


Cisco 642-889



Implementing Cisco Service Provider Next-Generation Edge Network Services

Version: 4.1

QUESTION NO: 1

Which type of VPN requires a full mesh of virtual circuits to provide optimal site-to-site connectivity?

- A. MPLS Layer 3 VPNs
- B. Layer 2 overlay VPNs
- C. GET VPNs
- D. peer-to-peer VPNs

Answer: B

Explanation:

QUESTION NO: 2

Which three Layer 3 VPN technologies are based on the overlay model? (Choose three.)

- A. ATM virtual circuits
- B. Frame Relay virtual circuits
- C. GRE/IPsec
- D. L2TPv3
- E. MPLS Layer 3 VPNs
- F. DMVPNs

Answer: C,D,F

Explanation:

QUESTION NO: 3

Which VPN technology uses the Group Domain of Interpretation as the keying protocol and IPsec for encryption that is often deployed over a private MPLS core network?

- A. DMVPN
- B. GET VPN
- C. SSL VPN
- D. L2TPv3

Answer: B

Explanation:

QUESTION NO: 4

What is the primary difference between 6PE and 6VPE?

- A. 6VPE does not require an MPLS core.
- B. 6VPE requires an IPv6-aware core.
- C. 6VPE provides IPv6 VPN services.
- D. 6VPE tunnels IPv6 packets inside IPv4 packets.

Answer: C

Explanation:

QUESTION NO: 5

Refer to the Cisco IOS XR router output exhibit,

```
RP/0/RP1/CPU0:R1#show route vrf red ipv6
```

Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP

i - ISIS, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, su - IS-IS summary null, * - candidate default

U - per-user static route, o - ODR, L - local

Gateway of last resort is not set

B 2001:db80:beef:1::/64

[200/0] via ::ffff:192.168.253.6 (nexthop in vrf default),07:04:14

which method is being used to transport IPv6 traffic over the service provider network?

- A. 6PE

- B. 6VPE
- C. native IPv6
- D. native IPv4
- E. dual stack

Answer: B

Explanation:

QUESTION NO: 6

Which flavor of MPLS Layer 3 VPN has MPLS enabled on PE-CE links?

- A. basic
- B. CSC
- C. inter-AS
- D. AToM
- E. VPLS

Answer: B

Explanation:

QUESTION NO: 7

Which MP-BGP address family must be configured to use VPLS autodiscovery in a Cisco IOS XR router?

- A. address-family l2vpn vpls-vpws
- B. address-family vpnv4 unicast
- C. address-family ipv4 mdt
- D. address-family ipv4 tunnel
- E. address-family vpls vfi

Answer: A

Explanation:

QUESTION NO: 8

In MPLS Layer 3 VPN implementations, what is used on the PE router to isolate potential overlapping routing information between different customers?

- A. route targets
- B. VRFs
- C. VC IDs
- D. pseudowire IDs
- E. pseudowire classes

Answer: B

Explanation:

QUESTION NO: 9

Within the service provider IP/MPLS core network, what must be implemented to enable Layer 3 MPLS VPN services?

- A. IS-IS or OSPF on all the PE and P routers
- B. MP-BGP between the PE routers
- C. RSVP on all the PE and P routers
- D. targeted LDP between the PE routers
- E. LDP between the CE and PE routers

Answer: B

Explanation:

QUESTION NO: 10

In MPLS Layer 3 VPN implementations, which mechanism is used to control which routes are imported to a VRF?

- A. RT
- B. RD
- C. VC ID
- D. PW ID
- E. VRF ID

Answer: A

Explanation:

QUESTION NO: 11

Which two statements regarding Cisco 6PE operations are correct? (Choose two.)

- A. The top label in the label stack is assigned by MP-BGP, and it is used to reach the egress PE.
- B. The inner label in the label stack is assigned by MP-BGP, and it is used for IPv6 forwarding at the egress PE.
- C. The top label in the label stack is assigned by LDP, and it is used to reach the egress PE.
- D. The inner label in the label stack is assigned by LDP, and it is used for IPv6 forwarding at the egress PE.

Answer: B,C

Explanation:

QUESTION NO: 12

Which Cisco IOS XR BGP configuration command is required to enable MP-BGP to carry IPv6 VPN routes?

- A. address-family ipv4 unicast
- B. address-family ipv6 unicast
- C. address-family vpnv4 unicast
- D. address-family vpnv6 unicast

Answer: D

Explanation:

QUESTION NO: 13

Which statement regarding the Cisco IOS BGP configuration exhibit is correct?

```
router bgp 65101
no bgp default ipv4-unicast
neighbor 172.16.1.1 remote-as 65101
neighbor 172.16.2.1 remote-as 65101
neighbor 172.16.3.1 remote-as 65101
!
address-family ipv4
neighbor 172.16.1.1 activate
neighbor 172.16.3.1 activate
!
address-family vpnv4
neighbor 172.16.2.1 activate
neighbor 172.16.3.1 activate
```

- A. None of the routers will receive IPv4 BGP routes.
- B. Only the 172.16.2.1 and 172.16.3.1 neighbors will receive both VPNv4 routes and IPv4 BGP routes.
- C. Only the 172.16.3.1 neighbor will receive both VPNv4 routes and IPv4 BGP routes.
- D. All three neighbors (172.16.1.1, 172.16.2.1, and 172.16.3.1) will receive both VPNv4 routes and IPv4 BGP routes.
- E. All three neighbors (172.16.1.1, 172.16.2.1, and 172.16.3.1) will receive IPv4 BGP routes.

Answer: C

Explanation:

QUESTION NO: 14

When implementing MPLS Layer 3 VPNs with customers running OSPF as the CE-PE routing protocol, the service provider MPLS backbone looks like what to the CE routers?

- A. the backbone (Area 0)
- B. an external routing domain
- C. a superbackbone that is transparent to the CE OSPF routers
- D. a transit area (similar to a transit area for supporting virtual links)

Answer: C

Explanation:

QUESTION NO: 15

When implementing MPLS Layer 3 VPNs with customers running OSPF as the CE-PE routing protocol, which situation will require a sham link to be implemented in the MPLS backbone?

- A. to connect customer sites in different OSPF areas
- B. to connect customer sites in the same OSPF area
- C. to prevent OSPF routing loops when a customer site has redundant CE-PE connections
- D. if there is a backdoor link between the CE routers, to ensure that the backdoor link is used only to back up the primary connection through the MPLS VPN

Answer: D

Explanation:

QUESTION NO: 16

On Cisco IOS and IOS XE Layer 3 MPLS VPN implementations, when redistributing the customer RIP routes into MP-BGP, the RIP metric is copied into which BGP attribute?

- A. local preference
- B. weight
- C. MED
- D. extended community

Answer: C

Explanation:

QUESTION NO: 17

Which option is used as a loop prevention mechanism to support MPLS VPN customers with multihomed sites?

- A. BGP down bit
- B. sham links
- C. AS override
- D. SOO extended BGP community
- E. allowas-in

Answer: D

Explanation:**QUESTION NO: 18**

Refer to the partial Cisco IOS XR PE router configuration exhibit for supporting a Layer 3 MPLS VPN customer using BGP as the CE-to-PE routing protocol.

```
router bgp 64500
address-family vpnv4 unicast
vrf Customer_A
address-family ipv4 unicast
!
neighbor 10.1.1.1
remote-as 64501
address-family ipv4 unicast
!
```

The service provider AS number is 64500, the customer AS number is 64501, and the customer CE router is 10.1.1.1. What is missing in the configuration?

- A. The route distinguisher has not been configured under router bgp 64500 vrf Customer_A.
- B. The import and export route targets have not been configured under router bgp 64500 vrf Customer_A.
- C. The 10.1.1.1 BGP neighbor has not been activated for IPv4 unicast routing.
- D. The 10.1.1.1 BGP neighbor has not been activated for the VPNv4 address family.

Answer: A

Explanation:

QUESTION NO: 19

Based on the Cisco IOS XR VRF configuration exhibit,

```
!  
vrf CustomerA-C  
description Customer A Central Site  
address-family ipv4 unicast  
import route-target  
1:210  
1:1000  
export route-target  
1:210  
1:1000  
!  
vrf CustomerB-C  
description Customer B Central Site  
address-family ipv4 unicast  
import route-target  
1:220  
1:1000  
export route-target  
1:220  
1:1000  
!  
vrf CustomerA  
description Customer A Site  
address-family ipv4 unicast  
import route-target  
1:210  
export route-target  
1:210  
!  
vrf CustomerB  
description Customer B Site  
address-family ipv4 unicast  
import route-target  
1:220  
export route-target  
1:220  
!
```

which two data flows between the MPLS VPNs will be allowed? (Choose two.)

- A. The CustomerA central site can communicate with the CustomerB central site.
- B. The CustomerA central site can communicate with all CustomerA sites.
- C. The CustomerA central site can communicate with all CustomerB sites.
- D. The CustomerA sites can communicate with all CustomerB sites.

Answer: A,B

Explanation:

QUESTION NO: 20

Refer to the partial Cisco IOS XR PE router VRF configuration exhibit.

```
vrf customer1
address-family ipv4 unicast
import route-target
 1:1
 2:1
export route-target
 1:1
 2:2
!
vrf customer2
address-family ipv4 unicast
import route-target
 1:2
 2:1
export route-target
 1:2
 2:2
!
```

To implement a central-service VPN supporting both customer1 and customer2, what will be the required corresponding VRF configuration on the central-service-server PE router?

A. vrf central-service-server
address-family ipv4 unicast
import route-target
3:1
2:2
export route-target
3:1
2:1
!

B. vrf central-service-server
address-family ipv4 unicast
import route-target

```
3:1
2:1
export route-target
3:1
2:2
!
C. vrf central-service-server
address-family ipv4 unicast
import route-target
3:1
1:1
1:2
export route-target
3:1
1:1
1:2
!
D. vrf central-service-server
address-family ipv4 unicast
import route-target
3:1
1:1
1:2
2:1
2:2
export route-target
3:1
1:1
1:2
2:1
2:2
!
```

Answer: A

Explanation:

QUESTION NO: 21

Which two statements about implementing a separate MPLS VPN to provide customers Internet access are correct? (Choose two.)

- A.** The Internet gateway router will act as a CE router.
- B.** Customers will use separate interfaces for VPN and Internet access.
- C.** Customers are assigned to the Internet VPN.

D. Internet routes will be leaked from the PE global routing table to the customer VRF.

Answer: A,C

Explanation:

QUESTION NO: 22

Which CE-to-PE routing protocol implements the down bit as a loop prevention mechanism?

- A. RIPv2
- B. EIGRP
- C. IS-IS
- D. OSPF
- E. BGP

Answer: D

Explanation:

QUESTION NO: 23

When you are using OSPF as the CE-to-PE routing protocol in MPLS VPN implementations, an OSPF route from customerA site 1 in Area 0 will appear as which kind of OSPF route in customerA site 2, also in Area 0?

- A. intra-area
- B. interarea
- C. E1
- D. E2

Answer: B

Explanation:

QUESTION NO: 24

In Layer 3 MPLS VPN implementations, if some of the VPNv4 routes on one PE router do not appear on another PE router, what could be the problem?

- A. RD mismatch between the PE routers

- B. RT export and import configuration errors
- C. RD export and import configuration errors
- D. VRF name mismatch between the PE routers

Answer: B

Explanation:

QUESTION NO: 25

When implementing CSC services, which two methods can be used to exchange label information between the downstream CSC customer carrier and the CSC backbone carrier? (Choose two.)

- A. using MP-BGP
- B. using RSVP
- C. using IGP and LDP
- D. using back-to-back VRF
- E. using front VRF and internal VRF

Answer: A,C

Explanation:

QUESTION NO: 26

The Cisco IOS XR address-family ipv4 labeled-unicast and the Cisco IOS/IOS XE neighbor send-label commands are used in which MPLS implementation?

- A. Cisco MPLS TE
- B. CSC using MP-BGP for label exchange
- C. back-to-back VRF
- D. AToM
- E. VPLS

Answer: B

Explanation:

QUESTION NO: 27

When verifying Layer 3 MPLS VPN operations, which Cisco IOS XR show command is best used to verify that the PE router is receiving the routes from the CE router?

- A. show route
- B. show route vrf vrf-name
- C. show bgp vpnv4 vrf vrf-name
- D. show bgp vpnv4 unicast ip-prefix

Answer: B

Explanation:

QUESTION NO: 28

When implementing Layer 3 MPLS VPNs on Cisco IOS/IOS XE PE routers, which PE-to-CE routing protocol requires a separate routing process to be created for each VRF?

- A. EIGRP
- B. RIPv2
- C. OSPF
- D. BGP

Answer: C

Explanation:

QUESTION NO: 29

In Layer 3 MPLS VPN implementations, if a customer is using the same AS number at both customer sites and the PE-to-CE routing protocol is BGP, what must be enabled on the PE router?

- A. BGP AS override
- B. BGP allowas-in
- C. BGP SOO extended community
- D. BGP AS path prepending

Answer: A

Explanation:

QUESTION NO: 30

Which three functions are performed by the PE router in an MPLS Layer 3 VPN? (Choose three.)

- A. exchanges routing updates with the CE router
- B. translates the CE routing information into VPNv4 routes
- C. exchanges VPNv4 routes with other PE routers over MP-BGP
- D. imports and exports RTs that are received from the P routers
- E. exchanges RDs with the P routers
- F. exchanges VPN labels with the CE routers

Answer: A,B,C

Explanation:

QUESTION NO: 31

Which BGP extended community is used to control the distribution of VPN routing information and to identify routers that may receive a set of routes that carry the community?

- A. SOO
- B. RT
- C. opaque
- D. route origin
- E. RD

Answer: B

Explanation:

QUESTION NO: 32

In which configuration mode is a route distinguisher configured in a Cisco IOS XR router?

- A. config-vrf
- B. config-vrf-af
- C. config-bgp
- D. config-bgp-af
- E. config-bgp-vrf

Answer: E

Explanation:

QUESTION NO: 33

Refer to the exhibit.

```
route-policy filter
  pass
end-policy
!
router bgp 1234
  bgp router-id 10.2.2.2
  address-family ipv4 unicast
    network 192.168.0.0/24
    network 192.168.1.0/24
  !
  neighbor-group share
    remote-as 1234
    update-source Loopback0
    address-family ipv4 unicast
      route-policy filter in
      route-reflector-client
    !
  !
  neighbor 10.1.1.1
    use neighbor-group share
  !
  !
vrf INTERNET
  rd 1:1
  address-family ipv4 unicast
    redistribute connected
  !
  !
  !
```

Given the output shown, which two statements are true? (Choose two.)

- A. The configured remote AS for neighbor 10.1.1.1 is 1234.
- B. Both prefixes that are referenced by network commands will be visible with the show bgp command from the information that is shown in the output.
- C. The neighbor 10.1.1.1 cannot learn any routes from this router.
- D. The router cannot learn any routes for neighbor 10.1.1.1.
- E. Routes from the Internet VRF that are injected into BGP through redistribution will be advertised to neighbor 10.1.1.1.

Answer: A,C

Explanation:

QUESTION NO: 34

Which Layer 2 VPN technology is implemented over an IP core network without the need for MPLS?

- A. VPLS
- B. VPWS
- C. AToM
- D. L2TPv3

Answer: D

Explanation:

QUESTION NO: 35

Which two Layer 2 VPN methods support interworking between customer sites with different Layer 2 encapsulation at each end (for example, Frame Relay to Ethernet interworking)? (Choose two.)

- A. AToM
- B. VPLS
- C. GET VPN
- D. L2TPv3

Answer: A,D

Explanation:

QUESTION NO: 36

What are the two AToM interworking modes? (Choose two.)

- A. bridged (interworking ethernet)
- B. routed (interworking ip)
- C. label-switched (interworking mpls)
- D. transparent (interworking transparent)

Answer: A,B

Explanation:

QUESTION NO: 37

When implementing EoMPLS on Cisco IOS XR routers, which command under the l2vpn configuration mode is used to define the pseudowire?

- A. pbb
- B. xconnect
- C. connect
- D. bridge
- E. bridge-domain

Answer: B

Explanation:

QUESTION NO: 38

When configuring an EoMPLS PW on a Cisco IOS XR router, what are the two supported transport modes? (Choose two.)

- A. ethernet (Ethernet port mode)
- B. ip (routed mode)
- C. vlan (VLAN-tagged mode)
- D. transparent (bridged mode)

Answer: A,C

Explanation:

QUESTION NO: 39

When implementing EoMPLS PWs, which configuration is optional?

- A. matching PW ID on both ends
- B. Xconnect
- C. Cisco Express Forwarding
- D. control word

Answer: D

Explanation:

QUESTION NO: 40

Which method is used to provide inter-AS AToM services?

- A. back-to-back VRF
- B. targeted LDP
- C. pseudowire stitching
- D. AToM interworking
- E. Cisco MPLS TE tunnels
- F. autodiscovery

Answer: C

Explanation:

QUESTION NO: 41

When troubleshooting EoMPLS configuration problems, which three parameters must match between the two ends of the pseudowire configurations? (Choose three.)

- A. control word usage
- B. MTU size
- C. pseudowire ID
- D. Xconnect group name
- E. EFP subinterface number

Answer: A,B,C

Explanation:

QUESTION NO: 42

Which Layer 2 protocol parameters can be carried inside the control word when implementing AToM service?

- A. PW ID
- B. Frame Relay FECN, BECN, and DE bits
- C. encapsulation type

D. VC type

Answer: B

Explanation:

QUESTION NO: 43

When implementing VPLS on Cisco routers, which data structure resembles a virtual switch and is used for learning the MAC addresses?

- A. VRF
- B. VFI
- C. SVI
- D. BVI

Answer: B

Explanation:

QUESTION NO: 44

In hierarchical VPLS implementations, which two access architectures can be used between the UPE and NPE? (Choose two.)

- A. EoMPLS
- B. MP-BGP
- C. Frame Relay
- D. 802.1ad

Answer: A,D

Explanation:

QUESTION NO: 45

Which two methods can be used for VPLS PW signaling? (Choose two.)

- A. static
- B. BGP
- C. IGP

- D. LDP
- E. RSVP

Answer: B,D

Explanation:

QUESTION NO: 46

When implementing nonhierarchical VPLS with eight PE routers, how many total PWs will be required between the PE routers?

- A. 8
- B. 16
- C. 28
- D. 32
- E. 64

Answer: C

Explanation:

QUESTION NO: 47

VPWS/EoMPLS offers which type of Ethernet services as defined by the MEF?

- A. E-Tree
- B. E-LAN
- C. E-Line
- D. E-Interworking

Answer: C

Explanation:

QUESTION NO: 48

When using the Cisco EVC software infrastructure, a double-tagged frame with a customer VLAN of 10 and a service provider VLAN of 150 will be best matched by which encapsulation configuration?

- A. encapsulation dot1q 10 second-dot1q any
- B. encapsulation dot1q 10 second-dot1q 150
- C. encapsulation dot1q 10 second-dot1q 50-200
- D. encapsulation dot1q 10
- E. encapsulation dot1q 150

Answer: B

Explanation:

QUESTION NO: 49

When implementing H-VPLS with QinQ access on Cisco Metro Ethernet switches, which two commands enable the QinQ tagging? (Choose two.)

- A. encapsulation dot1q {customer-vlan} second-tag {sp-vlan}
- B. encapsulation dot1q {sp-vlan} second-tag {customer-vlan}
- C. switchport mode dot1q-tunnel
- D. switchport mode trunk
- E. switchport access vlan {sp-vlan}
- F. switchport access vlan {customer-vlan}

Answer: C,E

Explanation:

QUESTION NO: 50

Implementing H-VPLS instead of VPLS reduces which requirement?

- A. having a full mesh of PWs between all the PE routers in the service provider MPLS core
- B. having a full mesh of PWs between all the UPE routers
- C. having to implement QinQ tagging between the UPE and the NPE
- D. having to implement MPLS LDP between the UPE and the NPE
- E. the overhead of using BGP or LDP autodiscovery

Answer: A

Explanation:

QUESTION NO: 51

When implementing VPLS on Cisco IOS XR routers, the customer-facing subinterfaces on the PE routers are assigned to which Cisco EVC component?

- A. bridge group
- B. bridge domain
- C. VFI
- D. Layer 2 transport
- E. BVI

Answer: B

Explanation:

QUESTION NO: 52

What is an advantage of using the Cisco EVC infrastructure to implement carrier-class Ethernet services that are not available on non-EVC-capable platforms?

- A. PW redundancy
- B. interworking support
- C. PW stitching support
- D. flexible frame-matching support and VLAN tag manipulation
- E. local cross-connect support

Answer: D

Explanation:

QUESTION NO: 53

When implementing a Layer 2 transport subinterface on a Cisco IOS XR router, which encapsulation option is used to match any packets that are not matched by any other service instances?

- A. default
- B. untagged
- C. any
- D. tag

Answer: A

Explanation:

QUESTION NO: 54

When configuring VPLS on the Cisco ASR 9000, which three configurations are required under the l2vpn configuration mode? (Choose three.)

- A. bridge-group
- B. bridge-domain
- C. xconnect
- D. vfi
- E. encapsulation

Answer: A,B,D

Explanation:

QUESTION NO: 55

Which option is the correct command to define an interface as Layer 2 on the Cisco ASR 9000?

- A. RP/0/RSP0/CPU0:R1(config)#int gigabitEthernet 0/6/0/0 l2transport
- B. RP/0/RSP0/CPU0:R1(config)#int gigabitEthernet 0/6/0/0 layer2
- C. RP/0/RSP0/CPU0:R1(config)#int gigabitEthernet 0/6/0/0 switchport
- D. RP/0/RSP0/CPU0:R1(config)#int gigabitEthernet 0/6/0/0 xconnect

Answer: A

Explanation:

QUESTION NO: 56

When implementing MPLS Layer 3 VPN services, which CE-PE routing method does not require the use of the redistribute command to enable the customer routes to be advertised through the MPLS cloud between the customer sites?

- A. EIGRP
- B. OSPF
- C. IS-IS
- D. BGP
- E. static routing
- F. OSPF or IS-IS

Answer: D

Explanation:

QUESTION NO: 57

In MPLS Layer 3 VPN implementations, what is used at the PEs to transform the customer IPv4 prefixes into a unique 96-bit prefix?

- A. RT
- B. RD
- C. VC ID
- D. PW ID
- E. AS number

Answer: B

Explanation:

QUESTION NO: 58

With Layer 3 MPLS VPN implementations on Cisco IOS XR PE routers, an interface is assigned to a VRF using the vrf command in which configuration mode?

- A. RP/0/RP0/CPU0:PE(config-vrf)#
- B. RP/0/RP0/CPU0:PE(config-if)#
- C. RP/0/RP0/CPU0:PE(config-bgp)#
- D. RP/0/RP0/CPU0:PE(config-bgp-af)#

Answer: B

Explanation:

QUESTION NO: 59

Refer to the partial Cisco IOS XR PE router configuration exhibit for supporting a Layer 3 MPLS VPN customer using EIGRP AS 20 as the CE-to-PE routing protocol.

```
router eigrp 10
vrf Customer_A
address-family ipv4
default-metric 10000 100 255 1 1500
autonomous-system 20
redistribute bgp 64500
interface GigabitEthernet0/0/0/0
!
router bgp 64500
vrf Customer_A
rd 64500:1
address-family ipv4 unicast
redistribute eigrp 10
!
```

The MPLS VPN customer is having problems receiving the EIGRP routes on the different customer site CE routers. What is wrong with this configuration that is causing the problem?

- A. The router eigrp command is referencing the wrong AS number.
- B. The redistribute eigrp command is missing the metric transparent option.
- C. The redistribute eigrp command is referencing the wrong AS number.
- D. The redistribute bgp command is missing the subnets option.
- E. The redistribute eigrp command is missing the subnets option.

Answer: C

Explanation:

QUESTION NO: 60

In which Cisco IOS XR configuration mode is the redistribute static command applied to enable the redistribution of static VRF routes between the PE routers?

- A. RP/0/RP0/CPU0:PE(config-router)#
- B. RP/0/RP0/CPU0:PE(config-bgp)#
- C. RP/0/RP0/CPU0:PE(config-bgp-vrf)#
- D. RP/0/RP0/CPU0:PE(config-bgp-vrf-af)#

Answer: D

Explanation:

QUESTION NO: 61

What is required on a Cisco IOS XR router to assign an interface to a VRF?

- A. Assign the VRF to the interface, and then re-enable Cisco Express Forwarding on the interface.
- B. Assign the VRF to the interface, and then reset the CE-to-PE routing protocol process.
- C. Shut the interface, assign the VRF to the interface, and then no shut the interface.
- D. Remove the IP address configuration on the interface, assign the VRF to the interface, and then reconfigure the IP address on the interface.

Answer: D

Explanation:

QUESTION NO: 62

When implementing VPLS on Cisco IOS XR routers, the VPLS PW neighbors can be statically defined under which configuration mode?

- A. bridge group
- B. bridge-domain
- C. vfi
- D. mpls ldp
- E. l2transport

Answer: C

Explanation:

QUESTION NO: 63

On Cisco IOS XR platforms using the EVC infrastructure, which command is used to enable a Layer 2 VPN subinterface?

- A. interface gi0/0/0/0.10
switchport access vlan 10
- B. interface gi0/0/0/0.10
switchport mode tunnel dot1q-tunnel
- C. interface gi0/0/0/0.10
switchport mode trunk
- D. interface gi0/0/0/0.10
bridge-group 10
- E. interface gi0/0/0/0.10 l2transport

Answer: E

Explanation:

QUESTION NO: 64

In Layer 3 MPLS VPN implementations, which protocol is used to carry the VPNv4 routes from PE to PE?

- A. RSVP
- B. IGP
- C. MP-BGP
- D. LDP

Answer: C

Explanation:

QUESTION NO: 65

What happens if the destination MAC address is not present in the table for the packets that are received on one of the ACs in VPLS?

- A. Packets are switched only to the PW.
- B. Packets are flooded only to the other local ACs.
- C. Packets are flooded on all other ACs and on all PWs that are associated with the bridge domain.
- D. Packets are dropped.

Answer: C

Explanation:

QUESTION NO: 66 DRAG DROP

Drag each item on the left to match the correct standard organization on the right	
VPWS and VPLS	IEEE
802.1ad and 802.1ah	IETF
E-Line, E-LAN, and E-Tree	MEF

Answer:

Drag each item on the left to match the correct standard organization on the right

VPWS and VPLS	802.1ad and 802.1ah
802.1ad and 802.1ah	VPWS and VPLS
E-Line, E-LAN, and E-Tree	E-Line, E-LAN, and E-Tree

Explanation: 802.1ad and 802.1ah ---- IEEE

VPWS and VPLS ---- IETF

E-Line , E-LAN and E-Tree --- MEF

QUESTION NO: 67 DRAG DROP

Drag the standard on the left to match the correct description on the right.

802.1ag	QinQ
802.1ad	PBB (also known as MAC-in-MAC)
802.1ah	Connectivity Fault Management

Answer:

Drag the standard on the left to match the correct description on the right.

802.1ag	802.1ad
802.1ad	802.1ah
802.1ah	802.1ag

Explanation: 802.1ah – PBB

802.1ad – QinQ

802.1ag – Connectivity fault management

QUESTION NO: 68 DRAG DROP

Match the LAN type on the left to the correct MEF service type on the right.

point-to-point	E-Tree
multipoint-to-multipoint	E-Line
rooted multipoint	E-LAN

Answer:

Match the LAN type on the left to the correct MEF service type on the right.

point-to-point	rooted multipoint
multipoint-to-multipoint	point-to-point
rooted multipoint	multipoint-to-multipoint

Explanation: Point-Point --- E-Line

Multipoint --to-Multipoint -- E-Lan

Rooted multipoint --- E-Tree

QUESTION NO: 69

Instructions

Enter the proper CLI commands and analysis the outputs on the Cisco routers to answer the multiple-choice questions.

From the network topology diagram, click on the router icon to gain access to the console of the router.

No console or enable passwords are required.

There are four multiple-choice questions with this task. Be sure to answer all four questions before selecting the Next button.

Not all the CLI commands or commands options are supported or required for this simulation.

For example, the show running-config command is **NOT** supported in this simulation.

All the devices in this simulation have been pre-configured and you are not required to enter in any configurations.

Scenario

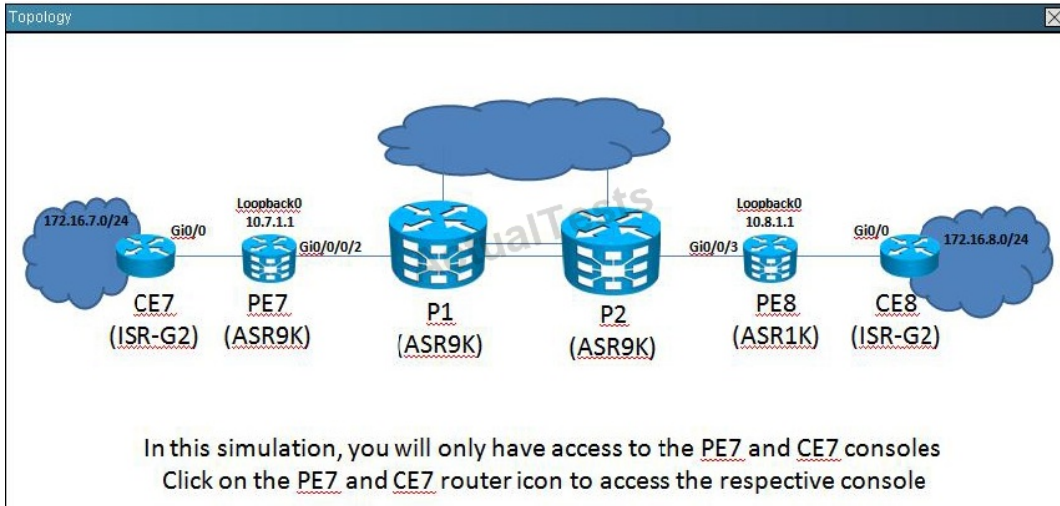
Referring to the network topology diagram shown in the exhibit, use the proper CLI commands on the CE7 and PE7 routers and interpret the supported CLI commands outputs to answer the four multiple choice questions.

The CE7 router is an ISR-G2 router and the PE7 router is an ASR9K router.

OSPF is the IGP running between all the PE and P routers and LDP is also running between all the PE and P routers.

The questions in this simulation are regarding the MPLS layer 3 VPN configurations on the PE routers where CE7

Scenario	Instructions	Topology	CE7	RP/0/RSP0/CPU0:PE7	Questions
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On PE7, which interface connects to the CE7 and what is the name of the VRF that interface is associated to? (Choose two.)

- A. Gi0/0/0/0
- B. Gi0/0/0/1
- C. Gi0/0/0/2
- D. Customer_1
- E. Customer_A
- F. Customer_CE7

Answer: B,C

Explanation: # show ip vrf interfaces

QUESTION NO: 70

Instructions

Enter the proper CLI commands and analysis the outputs on the Cisco routers to answer the multiple-choice questions.

From the network topology diagram, click on the router icon to gain access to the console of the router.

No console or enable passwords are required.

There are four multiple-choice questions with this task. Be sure to answer all four questions before selecting the Next button.

Not all the CLI commands or commands options are supported or required for this simulation.

For example, the show running-config command is **NOT** supported in this simulation.

All the devices in this simulation have been pre-configured and you are not required to enter in any configurations.

ActualTests

Scenario

Referring to the network topology diagram shown in the exhibit, use the proper CLI commands on the CE7 and PE7 routers and interpret the supported CLI commands outputs to answer the four multiple choice questions.

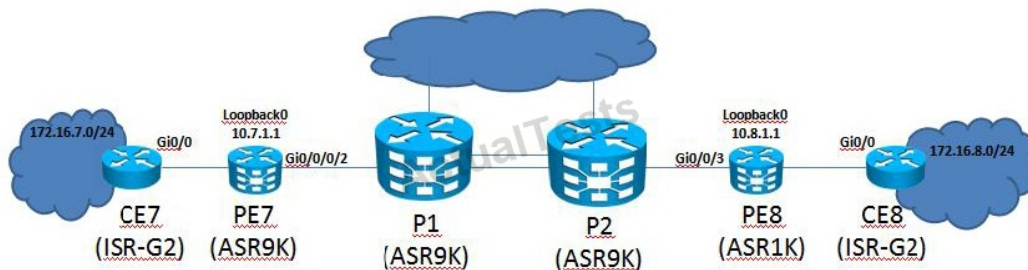
The CE7 router is an ISR-G2 router and the PE7 router is an ASR9K router.

OSPF is the IGP running between all the PE and P routers and LDP is also running between all the PE and P routers.

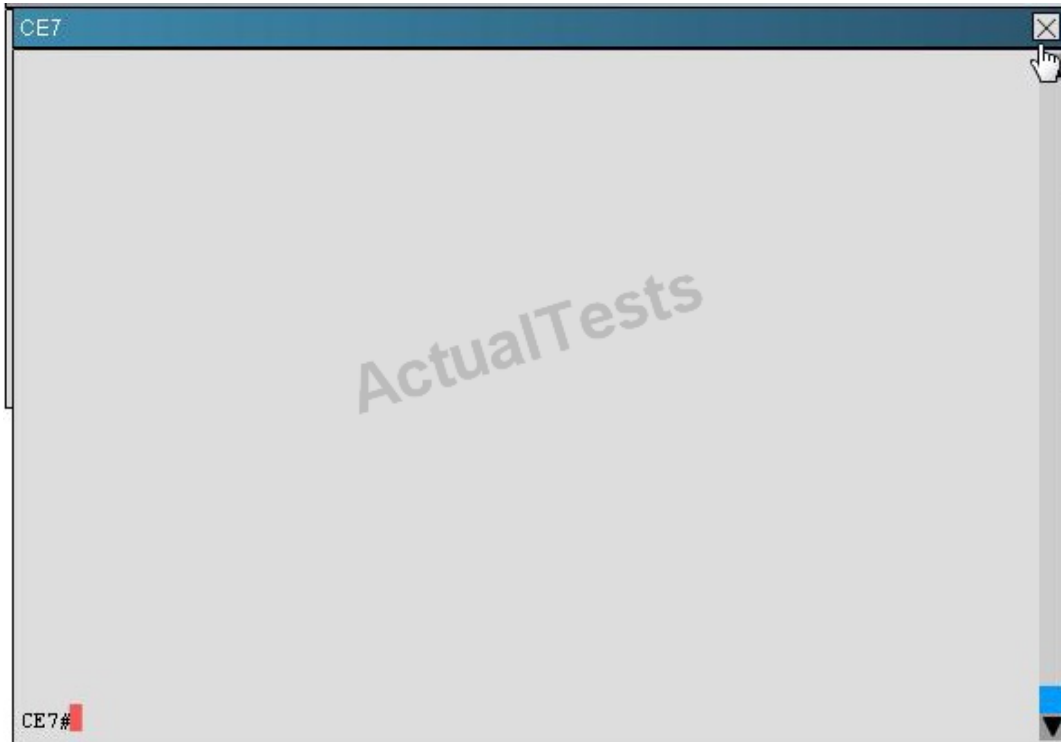
The questions in this simulation are regarding the MPLS layer 3 VPN configurations on the PE routers where CE7

Scenario	Instructions	Topology	CE7	RP/0/RSP0/CPU0:PE7	Questions
----------	--------------	----------	-----	--------------------	-----------

Topology



In this simulation, you will only have access to the PE7 and CE7 consoles
Click on the PE7 and CE7 router icon to access the respective console



What type of routing is used between CE7 and PE7?

- A. OSPF
- B. BGP
- C. is-is
- D. RIPv2
- E. Static routing

Answer: E**Explanation:** # show ip route**QUESTION NO: 71**

Instructions

Enter the proper CLI commands and analysis the outputs on the Cisco routers to answer the multiple-choice questions.

From the network topology diagram, click on the router icon to gain access to the console of the router.

No console or enable passwords are required.

There are four multiple-choice questions with this task. Be sure to answer all four questions before selecting the Next button.

Not all the CLI commands or commands options are supported or required for this simulation.

For example, the show running-config command is **NOT** supported in this simulation.

All the devices in this simulation have been pre-configured and you are not required to enter in any configurations.

ActualTests

Scenario

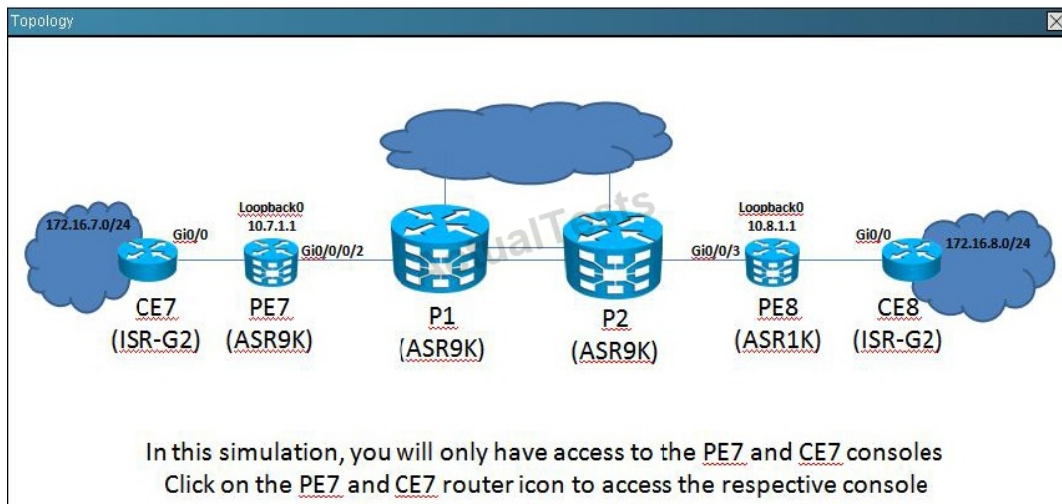
Referring to the network topology diagram shown in the exhibit, use the proper CLI commands on the CE7 and PE7 routers and interpret the supported CLI commands outputs to answer the four multiple choice questions.

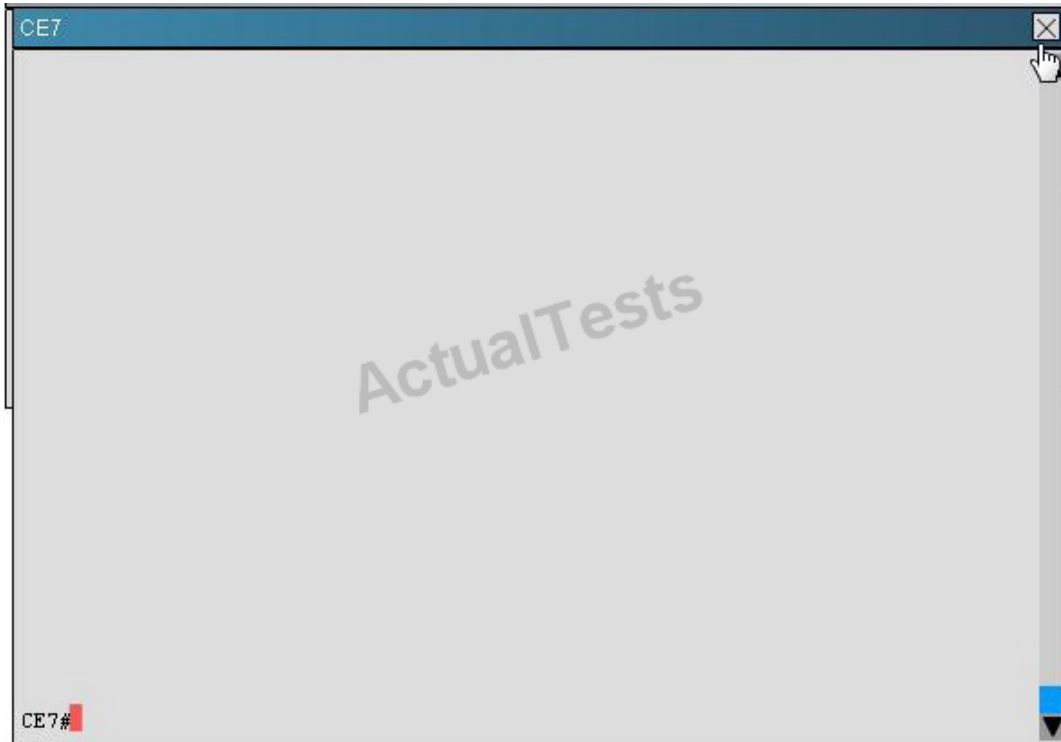
The CE7 router is an ISR-G2 router and the PE7 router is an ASR9K router.

OSPF is the IGP running between all the PE and P routers and LDP is also running between all the PE and P routers.

The questions in this simulation are regarding the MPLS layer 3 VPN configurations on the PE routers where CE7

Scenario	Instructions	Topology	CE7	RP/0/RSP0/CPU0:PE7	Questions
----------	--------------	----------	-----	--------------------	-----------





On PE7, how many multiprotocol IBGP routes are learned from PE8 and what is the next-hop IP address? (Choose two.)

- A. 1
- B. 2
- C. 3
- D. 10.8.1.1

- E. 172.16.8.1
F. 192.168.108.81

Answer: B,E

Explanation: show ip bgp vpnv4 all --- i tag field is the answer

QUESTION NO: 72

Instructions

Enter the proper CLI commands and analysis the outputs on the Cisco routers to answer the multiple-choice questions.

From the network topology diagram, click on the router icon to gain access to the console of the router.

No console or enable passwords are required.

There are four multiple-choice questions with this task. Be sure to answer all four questions before selecting the Next button.

Not all the CLI commands or commands options are supported or required for this simulation.

For example, the show running-config command is **NOT** supported in this simulation.

All the devices in this simulation have been pre-configured and you are not required to enter in any configurations.

Scenario

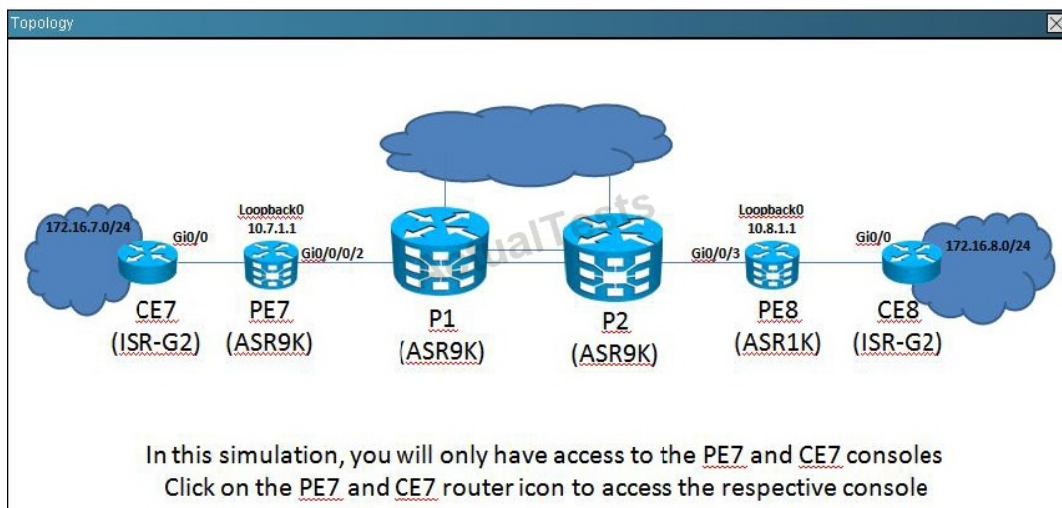
Referring to the network topology diagram shown in the exhibit, use the proper CLI commands on the CE7 and PE7 routers and interpret the supported CLI commands outputs to answer the four multiple choice questions.

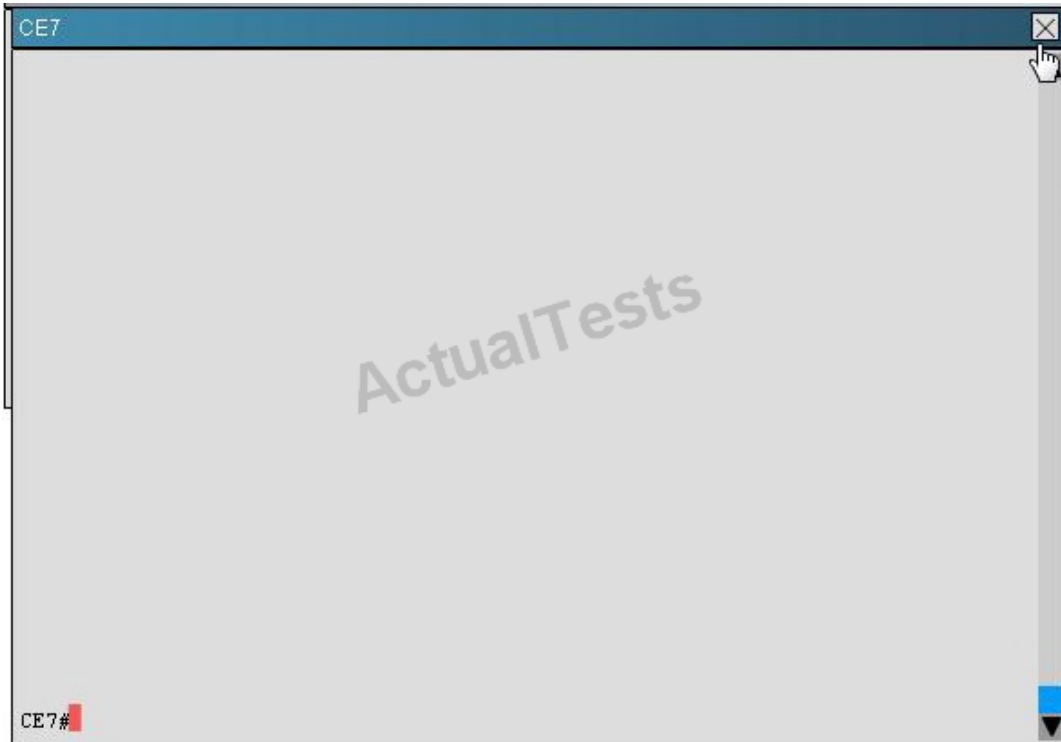
The CE7 router is an ISR-G2 router and the PE7 router is an ASR9K router.

OSPF is the IGP running between all the PE and P routers and LDP is also running between all the PE and P routers.

The questions in this simulation are regarding the MPLS layer 3 VPN configurations on the PE routers where CE7

Scenario	Instructions	Topology	CE7	RP/0/RSP0/CPU0:PE7	Questions
----------	--------------	----------	-----	--------------------	-----------





On PE7, which three statements are correct regarding the MPLS VPN configurations used to support the connectivity between the CE7 and CE8 sites?

(Choose three.)

- A. The RD is 1:1
- B. The import and export RTs are 1:1

- C. Interface Gi0/0/0/0 is associated to the "default" VRF
- D. The network that connects PE7 to CE7 is redistributed into multiprotocol IBGP
- E. The multiprotocol IBGP routes learned have a BGP origin code of "i"

Answer: B,C,E

Explanation:

```
# show ip route
show ip vrf
show ip vrf detail
```

QUESTION NO: 73

Instructions

Enter the proper CLI commands and analysis the outputs on the Cisco routers to answer the multiple-choice questions.

From the network topology diagram, click on the router icon to gain access to the console of the router.

No console or enable passwords are required.

There are four multiple-choice questions with this task. Be sure to answer all four questions before selecting the Next button.

Not all the CLI commands or commands options are supported or required for this simulation.

For example, the show running-config command is **NOT** supported in this simulation.

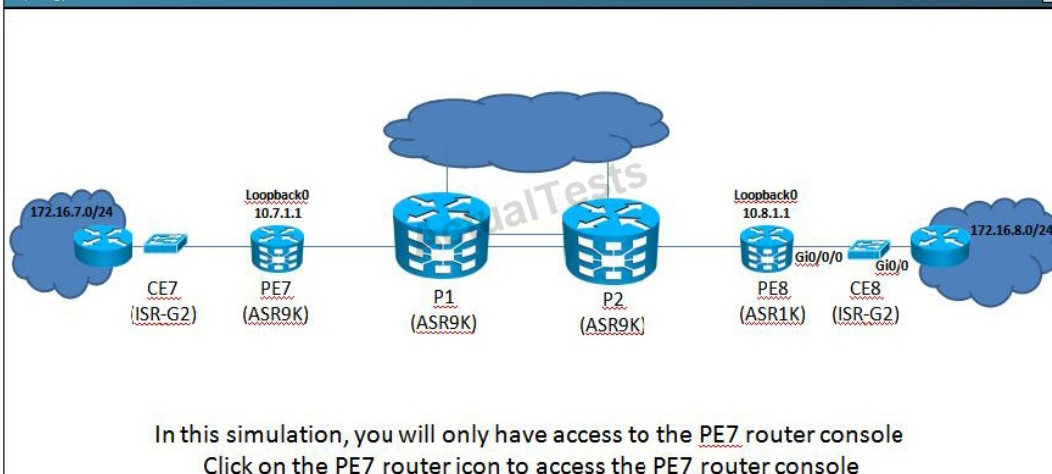
All the devices in this simulation have been pre-configured and you are not required to enter in any configurations.

Scenario

Referring to the network topology diagram shown in the exhibit, use the proper CLI commands on PE7 router and interpret the supported CLI commands outputs to answer the four multiple choice questions.

The PE7 router is an ASR9K router.

Topology





On PE7, what is the pseudowire ID that connects to the 10.8.1.1 neighbor?

- A. 70
- B. 80
- C. 123
- D. 0x840001
- E. 0x4000080

Answer: C

Explanation: # show vfi

QUESTION NO: 74

Instructions

Enter the proper CLI commands and analysis the outputs on the Cisco routers to answer the multiple-choice questions.

From the network topology diagram, click on the router icon to gain access to the console of the router.

No console or enable passwords are required.

There are four multiple-choice questions with this task. Be sure to answer all four questions before selecting the Next button.

Not all the CLI commands or commands options are supported or required for this simulation.

For example, the show running-config command is **NOT** supported in this simulation.

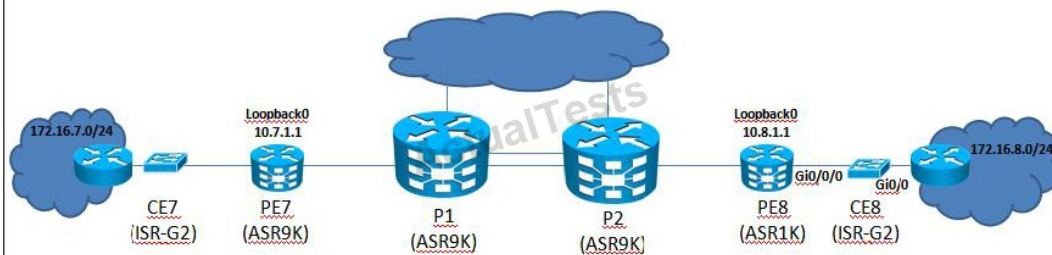
All the devices in this simulation have been pre-configured and you are not required to enter in any configurations.

Scenario

Referring to the network topology diagram shown in the exhibit, use the proper CLI commands on PE7 router and interpret the supported CLI commands outputs to answer the four multiple choice questions.

The PE7 router is an ASR9K router.

Topology



In this simulation, you will only have access to the PE7 router console
Click on the PE7 router icon to access the PE7 router console



Which statement is correct regarding the pseudowire on connects PE7 to the 10.8.1.1 neighbor?

- A. The control word is enabled on both ends of the pseudowire
- B. The MTU size is 1500 bytes on both ends of the pseudowire
- C. Pseudowire backup is enabled
- D. The pseudowire is in the down state

Answer: B

Explanation: show xconnect all

show ip interface brief

show mpls l2transport vc

show cable l2-vpn xconnect mpls-vc-map

QUESTION NO: 75

Instructions

Enter the proper CLI commands and analysis the outputs on the Cisco routers to answer the multiple-choice questions.

From the network topology diagram, click on the router icon to gain access to the console of the router.

No console or enable passwords are required.

There are four multiple-choice questions with this task. Be sure to answer all four questions before selecting the Next button.

Not all the CLI commands or commands options are supported or required for this simulation.

For example, the show running-config command is **NOT** supported in this simulation.

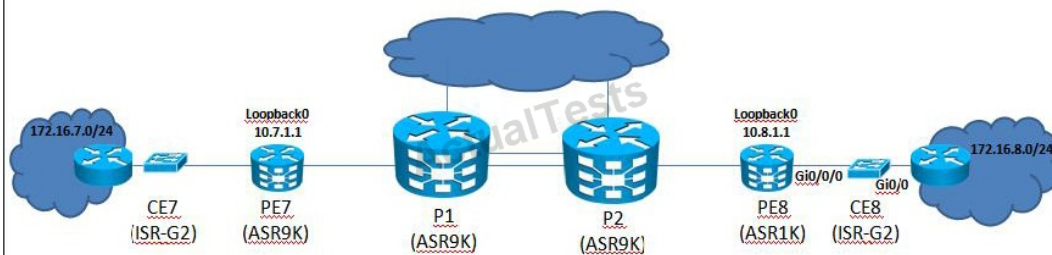
All the devices in this simulation have been pre-configured and you are not required to enter in any configurations.

Scenario

Referring to the network topology diagram shown in the exhibit, use the proper CLI commands on PE7 router and interpret the supported CLI commands outputs to answer the four multiple choice questions.

The PE7 router is an ASR9K router.

Topology



In this simulation, you will only have access to the PE7 router console
Click on the PE7 router icon to access the PE7 router console



On PE7, which encapsulation method is used on the pseudowire that connects to the 10.8.1.1 neighbor?

- A. MPLS
- B. L2TPv3
- C. IP
- D. LDP
- E. Ethernet

Answer: B

Explanation: show xconnect all

check value is mpls or l2tp or Ethernet etc in segment field

QUESTION NO: 76

Instructions

Enter the proper CLI commands and analysis the outputs on the Cisco routers to answer the multiple-choice questions.

From the network topology diagram, click on the router icon to gain access to the console of the router.

No console or enable passwords are required.

There are four multiple-choice questions with this task. Be sure to answer all four questions before selecting the Next button.

Not all the CLI commands or commands options are supported or required for this simulation.

For example, the show running-config command is **NOT** supported in this simulation.

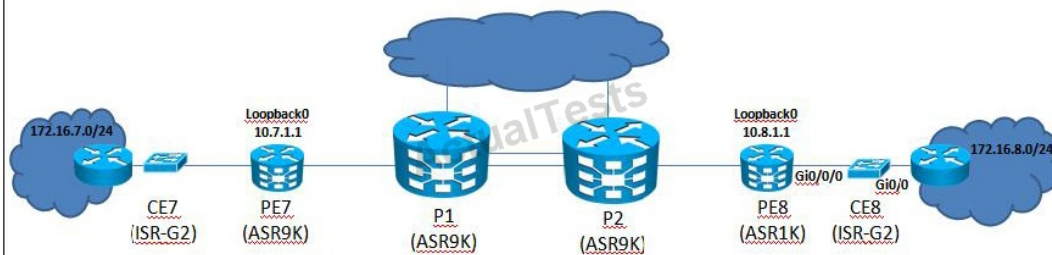
All the devices in this simulation have been pre-configured and you are not required to enter in any configurations.

Scenario

Referring to the network topology diagram shown in the exhibit, use the proper CLI commands on PE7 router and interpret the supported CLI commands outputs to answer the four multiple choice questions.

The PE7 router is an ASR9K router.

Topology



In this simulation, you will only have access to the PE7 router console
Click on the PE7 router icon to access the PE7 router console



On PE7, what is the PWtype on the p2p ac-pw named "testpw"?

- A. MPLS
- B. L2TPv3
- C. PPP
- D. LDP
- E. Ethernet

Answer: B

Explanation: show mpls l2transport vc testpw detail